

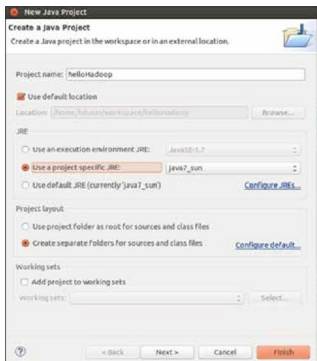


Figure 1: Creating a new Java project



Figure 2: Adding Hadoop libraries to a project

```
import org.apache.hadoop.fs.FSDataOutputStream;
import org.apache.hadoop.fs.Path;

public class helloWorld {
    public static final String theFilename = "hello.txt";
    public static final String message = "Hello, world!\n";

    public static void main (String [] args) throws
    IOException {

        Configuration conf = new Configuration();
        FileSystem fs = FileSystem.get(conf);

        Path filenamePath = new Path(theFilename);

        try {
            if (fs.exists(filenamePath)) {
                // remove the file first
                fs.delete(filenamePath);
            }

            FSDataOutputStream out = fs.create(filenamePath);
            out.writeUTF(message);
            out.close();

            FSDataInputStream in = fs.open(filenamePath);
            String messageIn = in.readUTF();
            System.out.print(messageIn);
            in.close();
        } catch (IOException ioe) {
```

```
            System.err.println("IOException during operation: " +
            ioe.toString());
            System.exit(1);
        }
    }
}
```

7. You must start the HDFS service before running this code.
8. You can check how successful the code is by running it. If, for some reason, your code does not work as intended, create a *jar* file and run it through Hadoop as *\$ hadoop jar /<your jar> helloWorld:*

```
$ bin/hadoop dfs -ls /user/hduser/
```

In this article, we learnt about HDFS and its usage as a general distributed file system. We found out how to use HDFS for our own application (apart from Hadoop applications), and for storing and accessing files where scalability, reliability and performance are critical to the application/service. HDFS can also be integrated with cloud services to provide storage facilities. Hence, it is an important package that will help you to learn and understand distributed systems. 

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